VendorFlow AI

Vendor onboarding, from submission to approval

AI Solutions Analyst case study, Problem Statement 2 (Operations)

Purpose	An automated process that takes a vendor submission and returns APPROVED, PENDING, or REJECTED with every reason visible, then tells the vendor exactly what is still needed.
Principle	AI reads documents and writes messages. Python makes every decision, so results are repeatable, testable, and easy to explain.
Status	Verification engine, decision rules, document extraction (Gemini plus offline fallback), message drafting, the pipeline runner, database, and reviewer rules and the Streamlit UI are built with 309 passing tests. Demo rehearsal, deployment, and the video remain.

1. Workflow

Every submission passes through seven stages. Each stage writes an event to the database as it happens, which drives the live run view and keeps a history even if a run fails part-way.

Stage	What happens	Done by	Build status
S1 Intake	Create a run ID, store the raw submission, hash every uploaded file, link to the parent run if this is a resubmission.	Python	**Done**
S2 Completeness	Check required fields and the tax document. List every gap, not just the first.	Python	**Done**
S3 Extraction	Read the PDF: document type, legal name, tax ID, signed, date. Text layer first, vision for scans. The model never sees the form.	pypdf + LLM	**Done**
S4 Grounding	Check what the model read is really in the document. Text PDFs: the value must appear in the PDF text. Scans: it must equal what the vendor typed.	Python	**Done**
S5 Verification	Run ten independent checks. Each records a result and its evidence. A crashed check becomes an error, never a pass.	Python	**Done**
S6 Decision	Any block gives REJECTED. Any other non-pass, including skipped, gives PENDING. Only all-pass gives APPROVED.	Python	**Done**
S7 Communication	Draft the vendor message from the safe request lines only. Internal evidence is never sent to the model.	LLM	**Done**
S8 Persistence	Save run, findings, extracted fields, decision, message, document hashes, and stage events.	SQLite	**Done**

After a decision. A PENDING run can be reviewed by a person (approve, reject, or request more information). The review is stored beside the system result and never overwrites it. When a vendor replies with evidence, the resubmission becomes a new run linked by *parent_run_id*, so the original findings stay intact, and the result shows what changed: findings resolved, still open, and new. A reviewer action writes its

own event (S9), so a run's status history can be rebuilt from events even though the run row is updated in place.

Severity rules

Finding	Meaning	Effect on status
pass	Check succeeded	No issue
warn	Additional evidence needed	PENDING
fail	Verification unresolved or incorrect	PENDING
error	Technical failure (not a verdict)	PENDING, never counted as a pass
skipped	Inputs missing, so the check could not run	PENDING; never APPROVED while any check is skipped
block	Confirmed disqualifying condition (match to a blocked vendor)	REJECTED

2. Verification checks

Check	Pass condition	If not passed
Required fields and document	All five fields and a tax document supplied	PENDING; every missing item listed in plain words
Tax ID format	Matches the country format (US EIN, UK VAT, Canada BN)	PENDING; unsupported country goes to manual review
Tax ID check digits	EIN prefix is on the IRS list; UK VAT passes mod 97 or the 9755 variant; Canadian BN passes Luhn	PENDING (most likely a typo); never REJECTED
Document type	W-9 (US), VAT certificate (UK), CRA letter (Canada)	PENDING; asks for the correct document
Signature and date	Signed (where required) and dated	PENDING; asks for a complete copy
Form vs document tax ID	Normalised IDs equal	PENDING with human review (may be a typo, may not)
Form vs document name	Core names match	PENDING; human review if very different; proof can clear it
Company vs bank holder name	Core-name similarity at least 90 and same entity type	60 to 89: PENDING, ask for DBA or parent proof. Below 60: PENDING with human review
Duplicate tax ID	No existing vendor uses it	PENDING with human review; REJECTED only if it belongs to a blocked vendor
Duplicate bank account	Not linked to a different vendor	PENDING with human review; REJECTED only if blocked
Document reuse	No other vendor's submission used the same file (by content hash); the resubmission chain is excluded	PENDING with human review

- **Grounding comes before the checks.** A hallucination only hurts if it produces a wrong pass, so a text value must appear in the PDF's text, and a scan's value must equal the form. This is valid only because the

model request never contains the form values (enforced by a test that captures the real request bodies).

- **Names are compared on their core.** Legal forms such as LLC, Inc, and Ltd are removed for scoring, so "Orion Components" matches "Orion Components LLC". A different legal form (LLC vs Inc) raises its own warning.
- **Duplicates are stored, not blocked.** The database has no unique constraints on tax ID or bank account, so duplicate submissions can be investigated.
- **Other vendors are never named** in anything sent to the vendor.

3. Tech stack

Layer	Choice	Reason
Language	Python 3.12	One language for logic, extraction, and UI
UI	Streamlit	Live run view and dashboard in the time available
Database	SQLite with SQLAlchemy	No setup; enough for synthetic demo data
Name matching	rapidfuzz	Fast, well-understood fuzzy scoring
PDF reading	pypdf, then LLM vision for scans	Cheap text extraction first; vision only when needed
AI model	Google Gemini free tier, with model failover (~20 requests/day per model)	Document extraction and vendor message wording only
Test data	reportlab and Pillow	Synthetic PDFs, including an image-only scan
Tests	pytest, pydantic	309 tests passing; typed contracts at the boundaries
Hosting	Streamlit Community Cloud	Live, runnable link for submission

4. Architecture

Presentation	streamlit_app.py + app_pages: Live run view | Audit dashboard | Review queue	**Done**
Orchestration	pipeline/runner.py: runs S1 to S8 and writes one stage event per step	**Done**
AI services	extract.py (S3 document reading) and message.py (S6 vendor wording). Offline fallback parses sample PDFs with pypdf.	**Done**
Business logic	checks.py (11 checks) | grounding.py | taxid.py (checksums) | decision.py (status) | names.py (matching) | config.py (policy) | review.py (reviewer rules)	**Done**
Data	SQLAlchemy models: vendors, runs, findings, documents, stage_events, reviews. Seeded with 4 existing vendors, one blocked.	**Done**

Data flows down the layers on a run and back up as stage events. The AI layer is deliberately thin: it returns structured fields and wording, and the business-logic layer decides what they mean.

5. Edge cases

Four are prepared for the live demo, alongside a clean happy path.

#	Scenario	What the process does	Result
1	**Parent or DBA mismatch, then resubmission.** Form says Orion Components LLC; bank account says Orion Components Holdings LLC.	Flags the partial name match and asks for DBA or parent proof. The vendor uploads an ownership certificate; a new run, linked to the first, clears the finding.	PENDING, then APPROVED
2	**Right EIN, wrong company.** Form and W-9 agree on the tax ID, but the W-9 names another company.	The tax ID check passes, but the document-name check fails and routes to human review. Tax ID alone would have approved it.	PENDING
3	**Bank account used by another vendor.**	Flags a possible ownership conflict for human review. The vendor message gives no detail about the other vendor. A reviewer can approve or reject with a note.	PENDING
4	**Wrong or unreadable attachment.** A utility bill instead of a W-9, or an unsigned scan.	Reports every problem found and asks for a signed, dated W-9 in plain words.	PENDING

Also covered by tests

- Three independent problems in one submission produce three findings (the engine never stops at the first failure).
- Blocked vendor's tax ID or bank account gives REJECTED. Unsupported country goes to manual review.
- Extraction failure or a crashed check is never treated as a pass.
- Reviewers cannot approve over a technical error or override a confirmed rejection.
- UK and Canada happy paths; tax-ID edge inputs such as lowercase and non-ASCII digits.

6. Assumptions

Area	Assumption
Countries	United States, United Kingdom, Canada. Others go to manual review.
Required documents	W-9, VAT certificate, CRA business number letter. These are design choices, not Zamp requirements.
Tax IDs	Format validation only; no registry lookup.
Bank accounts	No real bank verification.
Thresholds	Name similarity 90 (pass) and 60 (review) are configurable policy values.
Rejection	Only for confirmed disqualifying evidence: a match to a blocked vendor.
Human review	Required for unresolved discrepancies; stored separately from the system result.

7. Build plan

Day	Deliverable	Status
2	Database, sample PDFs, seed data, verification and decision engine, reviewer rules, tests	**Done**
3	Document extraction, message drafting, pipeline runner with stage events, resubmission linking	**Done**

Day	Deliverable	Status
4 to 5	Streamlit live run view, audit dashboard, review queue, demo presets	**Done**
6	Integration testing, deployment, demo rehearsal	**Not built**
7	Submit live link and a 5-minute demo video	**Not built**

Next: rehearse the four edge cases, deploy to a public URL, and record the 5-minute video.